

Preclinical advances and mechanistic insights of CAR-T therapy for acute myeloid leukemia: from target iteration to microenvironment regulation

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ABSTRACT

Introduction: Relapsed/refractory acute myeloid leukaemia (AML) carries a dismal prognosis, primarily due to profound biological heterogeneity and the scarcity of effective targeted therapies. Chimeric antigen receptor T (CAR-T) cell therapy has emerged as a transformative investigational strategy for AML by genetically engineering T cells to specifically target tumour antigens; however, its clinical translation is severely impeded by target antigen scarcity and the immunosuppressive tumour microenvironment (TME).

Discussion: This review systematically summarizes the preclinical advances and mechanistic underpinnings of CAR-T therapy for AML, focusing on two core bottlenecks. First, it elaborates on the screening of candidate antigens (e.g. CD33, CD123) and cutting-edge target optimization strategies, including nanobody-based CARs, logic-gated systems, adapter-CAR technology and combinatorial approaches to mitigate antigen escape. Second, it dissects the inhibitory effects of the AML TME on CAR-T function and reviews corresponding intervention tactics, such as immune checkpoint blockade, cytokine arming and gene editing. Integrating key updates from the 2024 ASH Annual Meeting, the review also highlights emerging preclinical technologies, including off-the-shelf CAR-T, CAR-NK cells and $\gamma\delta$ T cell therapy. Importantly, it acknowledges the prevalent preclinical-to-clinical translation gap, where promising lab efficacy has not yet translated into consistent clinical success.

Conclusions: Multidimensional technological innovation and the synergistic optimization of combination therapies are critical to overcoming AML-specific barriers. These advances hold the potential to unlock the precise clinical application of CAR-T therapy, ultimately improving survival outcomes for patients with relapsed/refractory AML.

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1. Introduction

AML, a haematological malignancy, is characterized by remarkable biological heterogeneity. Conventional chemotherapy and haematopoietic stem cell transplantation show limited efficacy in relapsed/refractory patients [1]. CAR-T therapy, which endows T cells with tumour antigen-specific recognition through genetic engineering, has emerged as a breakthrough approach for AML treatment [2]. However, inherent biological features of AML, such as the lack of tumour-specific antigens and the presence of an immunosuppressive tumour microenvironment, severely restrict the clinical application and development of CAR-T therapy in this field [3]. Additionally, clinical studies have shown that long-term and rigorous monitoring of patients receiving CAR-T immunotherapy is crucial, with particular vigilance for the potential progression risk of haematological malignancies [4].

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According to the 2023 report published by the European Society for Blood and Marrow Transplantation, the application scale of CAR-T therapy in the field of AML treatment continues to expand, fully demonstrating the significant advantages of the synergistic effect between cell therapy technology and traditional haematopoietic stem cell transplantation. This provides strong support for promoting the development of AML treatment towards precision immunotherapy [5]. Research data show that Switzerland has a leading application rate of CAR-T therapy among relevant countries, a phenomenon closely related to multiple factors such as the country's gross national income level, demographic age structure, and the distribution density of cell therapy centres [6].

CAR-T therapy, with its remarkable antitumour activity, has emerged as a highly promising innovative strategy in the field of AML treatment [7]. In this review, we extensively describe the development of CAR-T therapy for AML, mainly focusing on preclinical development and mechanistic aspects for improving CAR-T efficacy. To systematically clarify the preclinical value of this therapy, this review will conduct an in-depth exploration of the mechanism of action of CAR-T cells, target antigen screening for AML treatment, the regulatory role of the tumour immune microenvironment on CAR-T cell function, and emerging technologies in CAR-T therapy. Notably, this review emphasizes preclinical development and mechanistic investigations, whereas comprehensive coverage of clinical trial data, clinical efficacy endpoints and clinical translation barriers is limited. It is critical to acknowledge that impressive preclinical results do not always translate into clinical success – a key limitation that should be highlighted. The aim is to provide theoretical references for the standardized application and technological iteration of CAR-T therapy in AML treatment (Figure 1).

2. The mechanism of action of CAR-T cells

CAR-T cell therapy, a cutting-edge technology in the field of tumour immunotherapy, has brought new hope for conquering AML. By genetically engineering T cells, it endows T cells with the ability to precisely recognize and efficiently kill tumour cells, breaking through the limitations of traditional treatment methods and demonstrating unique therapeutic advantages and potential. In-depth analysis of the action mechanism of CAR-T cells is of great significance for optimizing treatment strategies, improving clinical efficacy and expanding the scope of application.

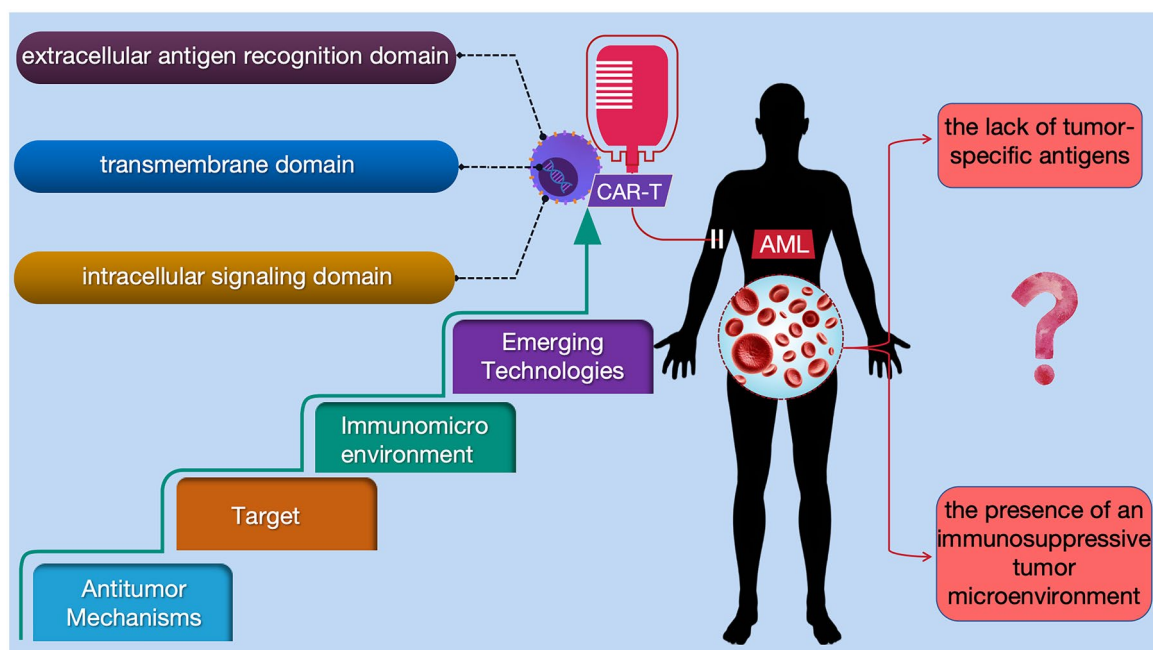


Figure 1. Overview of CAR-T therapy for AML: core mechanisms, key challenges and technical strategies.

2.1. The basic structure of CAR-T cells

The core function of CAR-T therapy relies on the engineered CAR structure, which is composed of three parts: the extracellular antigen recognition domain, transmembrane domain and intracellular signalling domain. Through synergistic effects, it endows T cells with the ability to specifically target and kill tumour cells (including antigen recognition, migration in the tumour microenvironment, and sustained survival) [8]. The structural design of CAR-T cells directly affects their targeting, activity and durability. In the treatment of AML, optimization should be performed according to the characteristics of antigen expression profiles and the tumour microenvironment, while remaining alert to the risk of clonal evolution potentially induced by long-term treatment [9].

The extracellular antigen recognition domain of CAR-T cells is typically a scFv, functioning as a 'scout' to specifically recognize and tightly bind to target antigens on the surface of AML cells. Common recognition targets include CD33, which is widely distributed in AML cells, and CLL-1, which is highly expressed in LSCs [10–12]. For instance, anti-CLL-1 CAR-T cells exhibit precise targeting of CLL-1 antigens on AML cell surfaces *via* their extracellular scFv, and activate T-cell signalling pathways through intracellular costimulatory domains (e.g. CD28 and OX40), significantly enhancing T-cell activation efficiency and *in vivo* persistence [13]. Moreover, CAR-T cells targeting HLA-DRB1 have demonstrated substantial potential in treating AML following allo-HSCT [14]. Studies on the monoclonal antibody KG2032 show that it can specifically bind to AML cells in approximately 50% of patients, primarily targeting phenotypes where the 86th amino acid in the HLA-DRB1 molecular subset is non-aspartic acid. It exhibits low cross-reactivity with normal leukocytes (except B lymphocytes) and non-haematopoietic tissues. KG2032-derived CAR-T cells have shown significant anti-leukaemia activity in preclinical models using female mice, holding promise as a novel therapeutic strategy for refractory AML patients who carry KG2032-reactive HLA-DRB1 alleles and receive allo-HSCT from KG2032-non-reactive donors.

The transmembrane domain of CAR-T cells, typically derived from CD28 or CD8 α , functions to stabilize the connection between extracellular and intracellular regions for efficient signal transduction. Studies have demonstrated that the CD8 α transmembrane domain enhances the antitumour activity of CAR-T cells by promoting inflammatory cytokine and granzyme B secretion, thereby improving the lysis of tumour cells with low-antigen expression levels. Compared with CD28-based CARs, CD8 α -containing CARs show superior *in vivo* antitumour efficacy [15]. Moreover, mutation of Fc receptor binding sites in the IgG1 CH2 domain of Hu8F4-CAR-T eliminates Fc γ receptor-mediated off-target effects and enhances targeting specificity [16]. *In vitro* and *in vivo* studies further confirm that optimized CAR-T cells specifically lyse HLA-A2+PR1+ leukaemia cells and effectively eliminate U937 cells in NSG xenograft models, demonstrating that transmembrane spacer optimization improves both the safety and antileukemic activity of CAR-T cells.

The intracellular signalling domain comprises costimulatory molecules (such as CD28, 4-1BB, CD27) and the CD3 ζ chain [17–20]. Among these, costimulatory molecules enhance T-cell activation, proliferation and durability – for instance, anti-CLL1 CAR-T cells containing 4-1BB or CD28/CD27 costimulatory domains exhibit comparable safety and efficacy in treating paediatric AML. The CD3 ζ chain, conversely, transmits the core signals for T-cell activation, driving effector function [21,22]. CAR-T cells of different generations differ structurally. First-generation CAR-T cells contain only the CD3 ζ signalling domain, yielding suboptimal activation. Studies have proposed replacing the traditional CD3 ζ domain with intracellular proximal T-cell signalling molecules (e.g. ZAP-70) to design the LINK CAR logic gating platform, which bypasses CD3 ζ to activate T cells, offering both potent antitumour activity and low toxicity [23]. The second generation incorporates costimulatory molecules like CD28 or 4-1BB to enhance activation efficiency. Research has confirmed that the CD28 CAR hinge exhibits local transient structures and conformational transitions within its overall disordered state, whose structural features may inform CAR design. CARs encoded by 4-1BB induce cell death by sequestering the ubiquitin-modifying enzyme A20 [24,25]. The third generation includes two costimulatory domains, further optimizing function. Studies have validated that third-generation CARs (CD28-4-1BB-CD3- ζ) based on biotin-binding monomeric streptavidin enable highly specific engineered T cells for potential tumour immunotherapy [26].

Notably, the design of CAR structures requires a balance between targeting specificity and safety. For example, anti-CD19 therapies (including CAR-T cells and blinatumab) may promote the expansion of specific mutant clones through selective pressure, even leading to lineage switch of B-lineage ALL to AML (KMT2A wild-type), which highlights the need to consider antigen-targeting precision and potential lineage transformation risks in AML-related CAR design [27].

2.2. Technical breakthroughs and risk assessment in CAR-T cell therapy research

In the research of CAR-T cell therapy targeting AML, the synergistic breakthrough of technological innovation and risk management represents the core direction for improving clinical translation efficiency [28].

To accurately evaluate the efficacy of CAR-T cells against AML, bioluminescent AML patient-derived xenograft models constructed in mice play a crucial role. These models precisely recapitulate the complexity and heterogeneity of patient samples, providing experimental conditions that highly simulate the human environment for therapy evaluation [29]. However, traditional cell therapy evaluation technologies have limitations: the luciferase-substrate combinations commonly used in bioluminescent imaging struggle to monitor dual cell populations simultaneously, leading to cumbersome evaluation procedures and high costs. To address this, researchers developed a dual luciferase system by combining the green click beetle luciferase CBG99 with the red firefly luciferase mutant Akaluc [30]. Validated through pairing of CAR-T cells and tumour cells, this system enables real-time tracking of dual cells not only in 2D/3D *in vitro* culture environments but also in *in vivo* models, and is compatible with multiple types of detection equipment. This not only greatly facilitates studies on CAR-T-tumour cell interactions but also provides an efficient tool for other research involving dual-cell component monitoring.

From the perspective of risk prediction, CAR-T therapy is associated with the risk of triggering therapy-related myeloid neoplasms. However, the clonal haematopoiesis risk score now enables specific prediction to identify high-risk patients in advance, allowing clinicians to formulate more targeted preventive and therapeutic strategies [31]. Additionally, the modified EASIX score can predict severe CRS/ICANS in AML patients following CLL1 CAR-T cell therapy, providing a basis for clinicians to intervene promptly and reduce the harm of adverse reactions [32].

In the technical aspect of CAR-T cell preparation, an adequate absolute lymphocyte count (ALC) is a necessary prerequisite for autologous CAR-T cell production. Although the intensity of chemotherapy in AML patients affects ALC recovery, causing its level to decline with the progression of chemotherapy courses, most patients still meet the requirements for apheresis before each course. This suggests that clinically, early collection of T cells can be used to secure CAR-T treatment opportunities for high-risk relapsed and refractory patients [33]. Moreover, to overcome the efficiency bottleneck of non-viral CAR-T cell production, studies have applied the piggyBac transposon combined with electroporation technology, supplemented with allogeneic feeder cells, to successfully generate CD19- and CD123-specific CAR-T cells compliant with GMP standards, strongly promoting the development of CAR-T cell preparation technology [34]. Additionally, CAR-T cells targeting IL-1RAP on the surface of leukemic stem cells have been developed for AML, and their preclinical production processes have been validated through the CliniMACS® Prodigy semi-automated closed system in compliance with Good Manufacturing Practice standards. This has laid a solid foundation for their entry into phase I clinical trials, holding promise to bring new therapeutic hope to AML patients [35].

2.3. Antitumour mechanisms of CAR-T cells

2.3.1. Target recognition and binding

In the research of CAR-T cell therapy, antigen-targeted recognition and binding represent the core mechanism by which CAR-T cells exert antitumour activity. The scFv on the surface of CAR-T cells specifically recognizes and binds to antigens on tumour cell surfaces, inducing immunological synapse formation. This process triggers the cascade activation of intracellular signalling pathways in CAR-T cells, promoting the release of cytotoxic effector molecules (such as perforin and granzyme) and driving CAR-T cell proliferation, ultimately enabling efficient and precise killing of tumour cells [36]. In recent years, immunological synapse biosensors constructed based on FRET technology have become important tools for predicting CAR-T cell function [37]. These sensors enable real-time, dynamic monitoring of the formation efficiency of immunological synapses and the intensity of signal transduction during the interaction between CAR-T cells and tumour cells, providing critical data support for dissecting the functional regulatory mechanisms of CAR-T cells. By integrating the detection results of FRET biosensors, researchers can optimize CAR structure design and precisely regulate the antigen recognition specificity and killing

activity of CAR-T cells, thus advancing the development of next-generation CAR-T cell therapies with stronger antitumour efficacy.

To address the challenge of lacking safe and effective targets for CAR-T cells in AML, researchers have identified CSF1R and CD86 as novel targets based on high-resolution transcriptomic maps constructed using single-cell RNA sequencing technology [38]. CAR-T cells specifically recognize CSF1R or CD86 antigens on tumour cell surfaces through their surface scFv, forming immunological synapses that activate intracellular signalling pathways in CAR-T cells. *In vivo* studies have confirmed that CSF1R/CD86-targeted CAR-T cells can efficiently recognize tumour cells, exhibit significant antitumour activity, and show no obvious off-target toxicity, providing a new direction for research on CAR-T targeting mechanisms in AML.

To address the off-target toxicity caused by traditional targets (such as CD123 and CD33) co-expressed in normal haematopoietic and endothelial cells, researchers have developed a dual CAR strategy. By co-expressing the low-affinity anti-CD123 IL-3-zetakine and the activation signal-free anti-CD33 costimulatory receptor (CD33.CCR) in CIK, CAR-T cells can precisely target CD123⁺/CD33⁺ leukaemia cells through scFv-mediated specific recognition of dual antigens. This approach avoids toxic damage to haematopoietic stem/progenitor cells and endothelial cells, providing an innovative solution for optimizing the target recognition specificity of CAR-T cells [39].

Based on the characteristic that B7-H3 is highly expressed in AML cells but lowly expressed in normal tissues, researchers have constructed B7-H3 CAR-T cells [40]. In this system, the scFv of CAR-T cells specifically binds to B7-H3 antigens on the surface of tumour cells, forming immunological synapses that trigger CAR-T cell activation. *In vivo* studies have shown that these cells can significantly inhibit tumour growth in AML mouse models and prolong survival. These findings reveal the effectiveness of B7-H3 as a CAR-T cell target and exemplify the core antitumour mechanism by which CAR-T cells recognize and bind to tumour cells through targeted antigens (B7-H3).

To address the collateral damage to normal myeloid cells in AML treatment, researchers have constructed a split dual CAR system based on 'AND gate logic', which consists of an NKG2D extracellular domain (linked to DAP12) and a PD-L1-specific costimulatory receptor (containing 4-1BB). CAR-T cells are activated only when their surface scFvs simultaneously recognize both NKG2D ligands and PD-L1 antigens and form immunological synapses. This system selectively targets cells co-expressing NKG2D ligands and PD-L1 (such as AML cells and LPS-polarized macrophages), demonstrating higher selectivity for normal myeloid cells. *In vivo* studies show it can effectively eradicate KG-1 AML xenografts, exemplifying the mechanism by which CAR-T cells achieve precise target recognition through dual-antigen pairing [41].

In the target optimization strategies for CAR-T therapy in AML, new achievements have been made in the field of dual-target CAR-T. Researchers have developed dual-target CD33-CLL1 CAR-T cells, which contain bispecific single-chain variable fragments [42]. The two scFvs on the surface of these CAR-T cells recognize CD33 and CLL1 antigens respectively, forming immunological synapses with tumour cells to activate cytotoxic functions. *In vivo* models demonstrate its significant antileukemic activity, capable of eradicating tumours and prolonging survival. This strategy represents an effective and safe approach for AML treatment, holding promising potential for clinical application.

2.3.2. T cell activation and proliferation

In CAR-T cell therapy for AML, T cell activation and proliferation represent core processes determining treatment efficacy, with studies on their molecular mechanisms and optimization strategies being crucial for enhancing therapeutic outcomes. From the perspective of signalling pathway regulation, the CD3 ζ and costimulatory signal modules in the CAR structure synergistically activate downstream key pathways such as NF- κ B and NFAT, driving CAR-T cell expansion and establishing a cellular quantity basis for anti-tumour immune responses [43,44]. Research further indicates that the binding affinity of CAR molecules and the intensity of tonic signalling play key roles in T cell function optimization [45]. Taking anti-Her2 CAR as an example, amino acid sequence modification of the scFv can significantly enhance its antigen-specific binding capacity, cell affinity, and 'nourishing' signal intensity. Specifically, a strategy of substituting highly hydrophobic amino acids within the scFv can promote CAR antigen affinity, enhance nonspecific binding, increase receptor aggregation and improve tonic signalling efficiency by strengthening intermolecular hydrophobic interactions. This in turn supports T cell function maintenance, ensuring their antitumour activity during antigen restimulation and long-term tumour control.

To address the dilemma of scarce tumour-specific antigens in AML treatment, a research team has innovatively constructed a dual-receptor T cell platform composed of an AbTCR and a chimeric CSR [46]. The platform precisely targets the WT1 RMFPNAPYL epitope/HLA-A2 complex *via* AbTCR, providing the first signal for T cell activation. Concurrently, CSR uses a CD33 scFv fused with a truncated CD28 fragment to deliver the critical costimulatory second signal. Through the synergistic action of the dual receptors, this platform enables efficient T cell activation and proliferation, endows specific killing capacity against AML cells, and selectively spares normal haematopoietic cells such as CD33⁺ bone marrow monocytes. This approach optimizes T cell activation and proliferation mechanisms while effectively reducing treatment-related toxicity.

At the metabolic regulation level, AML leukemic cells enhance glycolysis (Warburg effect) to competitively uptake key nutrients such as glucose from the microenvironment, leading to functional impairment and proliferative damage of CAR-T cells due to nutrient deprivation [47]. This is specifically characterized by downregulation of key molecules including GLUT1 and mitochondrial TFAM, thereby inhibiting mitochondrial biogenesis. In response, researchers employed lentivirus-mediated gene transfection technology to achieve overexpression of SLC2A1 and/or TFAM genes in T cells. Through metabolic reprogramming, this approach enhances glucose uptake capacity of T cells, promotes IL-2 release, reduces cellular exhaustion, and effectively suppresses leukaemia progression *via* a dual action mechanism of 'nutrient competition+cytotoxicity'. This provides a novel theoretical basis for metabolic regulation of CAR-T cell activation and proliferation.

Additionally, in terms of signalling pathway intervention and biomarker discovery, the Siglec6 CAR-T cell therapy targets the SHP2/Src/ERK/IL-3 signalling axis, effectively inhibiting Src and ERK signalling activation induced by Siglec6 and SHP2. This significantly reduces the proliferation, invasion and migration capabilities of AML cells. Compared with traditional CD19 CAR-T therapy, it can induce a higher level of AML cell apoptosis and reduce their distribution in mice, thus effectively curbing AML progression [48]. Meanwhile, studies have confirmed that ADGRG1 can serve as a specific biomarker for CD8⁺ tumour-reactive T cells, exhibiting selectively upregulated characteristics during antigen-TCR interaction. Moreover, ADGRG1⁺ CD8⁺ T cells demonstrate stronger IFN- γ secretion capacity and cell killing function when contacting matched primary AML cells [49].

Notably, the AML microenvironment exerts significant inhibitory effects on CAR-T cell function [50]. Studies have shown that myeloid leukaemia cells can induce downregulation of CAR molecule expression on the surface of CAR-T cells under conditions of low tumour cell-to-T cell ratios, thereby inhibiting T cell activation and cytotoxicity. Mechanistic investigations demonstrate that leukaemia-derived Galectin-1 serves as a key molecule mediating this process. Based on these findings, strategies such as inhibiting lysosomal degradation or actin polymerization effectively alleviate CAR downregulation and restore T cell function. Additionally, mutational design of the CAR cytoplasmic region (e.g. CD33 CARKR) by substituting lysine with arginine significantly reduces CAR downregulation and enhances T cell killing activity. The application of Galectin-1-specific inhibitory peptide (anginex) also rescues T cell activation defects effectively. These discoveries reveal a novel mechanism of T cell function suppression in the AML microenvironment, providing highly promising intervention targets for optimizing CAR-T cell activation and proliferation strategies.

2.3.3. Tumour cell killing

The killing effect of CAR-T cells on AML cells is a complex biological process mediated by multi-mechanistic collaboration, primarily involving classical immune effector pathways and metabolic regulatory networks. The interaction of these mechanisms collectively constitutes the molecular basis of their anti-tumour activity.

The perforin/granzyme pathway plays a central killing role in classical immune effector pathways [51]. Upon activation *via* antigen-specific recognition, CAR-T cells directly release perforin, which polymerizes to form transmembrane pores on the AML cell membrane, facilitating the entry of granzymes into tumour cells. Once inside the cells, granzymes activate the caspase cascade, inducing programmed cell death of target cells. This pathway serves as the direct effector mechanism for CD7 CAR-T and other therapies to eliminate CD7-positive AML cells, providing a fundamental basis for tumour cell killing [52].

The perforin/granzyme pathway serves as the primary effector mechanism by which CAR-T cells directly induce apoptosis of AML cells. The death receptor pathway, represented by the Fas/FasL signalling axis, acts

as a critical complementary and synergistic pathway in tumour cell killing [53]. Specifically, FasL on CAR-T cells binds to Fas receptors on AML cells, activating the caspase-8-mediated apoptotic signalling cascade. Notably, in TP53-mutated AML cells, the Fas/FasL pathway collaborates with the perforin/granzyme pathway to effectively compensate for the killing resistance caused by TP53 mutations. Studies show that interaction between TP53-deficient AML cells and CAR-T cells prolongs CAR-T cell action time and upregulates exhaustion markers, thereby reducing killing efficiency [54]. Transcriptomic analysis reveals that during this process, the mevalonate pathway is aberrantly upregulated and the Wnt pathway is significantly downregulated in AML cells. Targeted regulation of these two signalling pathways can effectively restore AML cells' sensitivity to CAR-T cell killing, indicating that the mevalonate pathway is a key target for overcoming TP53 defect-related drug resistance, while modulation of the Wnt pathway helps enhance therapeutic efficacy.

Cytokine-mediated immunomodulation is indispensable in the anti-tumour process of CAR-T cells. IFN- γ secreted by CAR-T cells activates innate immune cells such as macrophages and NK cells in the tumour microenvironment, enhancing anti-tumour immune responses *via* ADCC [55]. IFN- γ not only enhances the killing activity of endogenous T cells and NK cells but is also crucial for maintaining the cytotoxicity of CAR-T cells themselves [56]. Meanwhile, IFN- γ produced by CAR-T cells promotes host secretion of IL-12, forming a positive immunoregulatory circuit that further supports host immune responses and the anti-tumour activity of CAR-T cells [57]. Additionally, PD-L1 blockade therapy can restore the killing activity of CAR-T cells by regulating CD163+ M2-type macrophages *via* IFN- γ [58]. Studies have confirmed that the intrinsic sensitivity of tumour cells to the pro-apoptotic effect of IFN- γ is a key factor determining the anti-tumour activity of CD4+ CAR-T cells [59].

The glucose transporter GLUT5 provides a novel metabolic mechanism for the survival and functional performance of CAR-T cells in the AML tumour microenvironment [60]. GLUT5 specifically transports fructose. Under conditions of high fructose and glucose-free tumour microenvironments, GLUT5 restores the glycolytic capacity of CAR-T cells and promotes ATP production. $^{13}\text{C}_6$ -labeled fructose tracing technology confirms that GLUT5-mediated fructose metabolism supports glycolytic reprogramming and tricarboxylic acid cycle repair in CAR-T cells. Additionally, GLUT5 significantly enhances the migration speed of CAR-T cells, strengthens their 'serial killing' effect, and effectively prolongs the anti-tumour action time of CAR-T cells in AML xenograft models. This offers a new metabolic engineering strategy to overcome CAR-T cell functional exhaustion caused by glucose deprivation in the tumour microenvironment.

Furthermore, to enhance the therapeutic potency and persistence of CAR-T cells in AML, methodological strategies for improving T-cell heterogeneity within the graft warrant comprehensive consideration. The optimal timing of peripheral blood T-cell collection directly determines the viability, differentiation status, and antitumour potential of final CAR-T products; early collection prior to intensive salvage chemotherapy is recommended to preserve the proportion of naive and central memory T-cell subsets, thereby avoiding T-cell dysfunction and exhaustion induced by myelosuppression and tumour microenvironment exposure. Additionally, prior chemotherapeutic regimens exert profound impacts on the quality of collected T cells: intensive cytotoxic chemotherapy and alkylating agents may cause persistent T-cell repertoire contraction, DNA damage, and reduced proliferative capacity, whereas low-intensity chemotherapy or hypomethylating agents have relatively mild effects on T-cell heterogeneity and function. Rational combination treatment approaches, such as sequential administration of low-dose cytokines (IL-7, IL-15) during T-cell enrichment and culture, can effectively expand memory T-cell subsets and maintain phenotypic diversity, further elevating the heterogeneity and functional robustness of CAR-T products.

3. Target selection for CAR-T therapy in AML

In the field of haematological malignancy treatment, CAR-T therapy has become a critical therapeutic modality for B-cell malignancies, demonstrating remarkable clinical efficacy. However, constrained by the lack of tumour targets with high specificity and optimal safety, CAR-T therapy has not yet achieved significant breakthroughs in AML treatment. In this regard, in-depth exploration of potential target antigens for CAR-T therapy in AML and systematic collation of current target optimization strategies are of great significance for advancing the development of CAR-T technology in AML treatment (Figure 2).

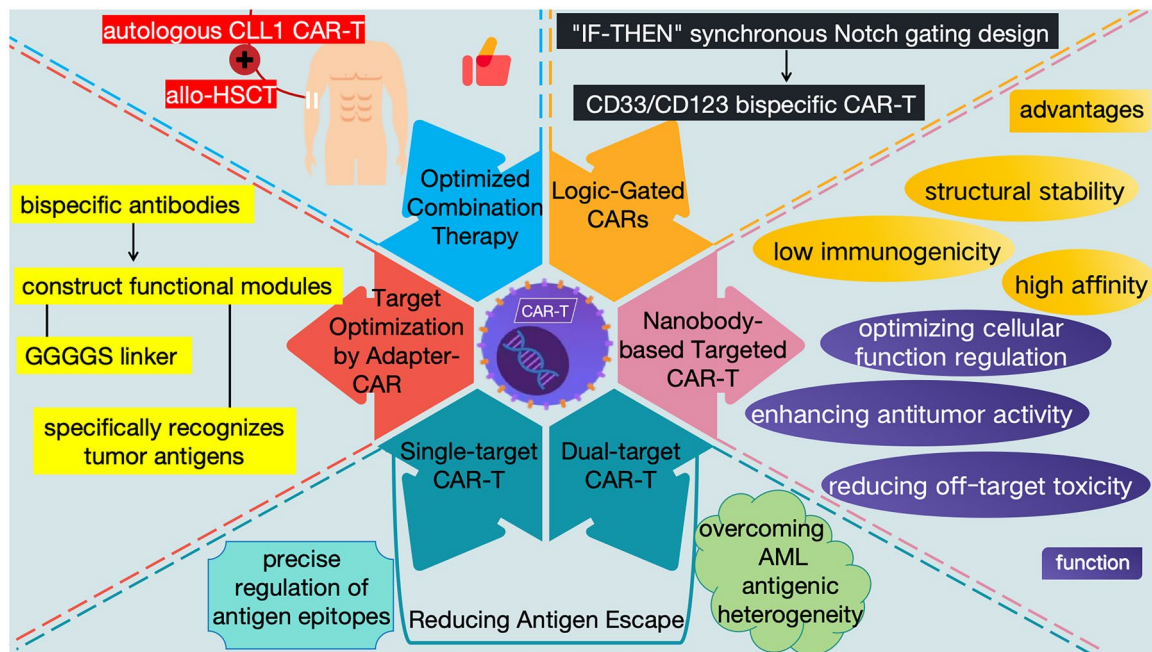


Figure 2. Key target antigens and optimization strategies for AML CAR-T therapy.

3.1. Potential target antigens

The lack of highly specific cell surface markers in AML severely restricts the clinical application expansion of CAR-T cell therapy. Through ‘structural surface proteomics’ technology (cross-linking mass spectrometry combined with glycoprotein surface capture), studies have successfully identified the activated conformation of integrin $\beta 2$ as a potential AML-specific target antigen. CAR-T cells developed based on this target have demonstrated significant AML cell clearance capacity in xenograft models, with low toxicity to normal haematopoietic cells. This discovery provides a novel potential target antigen for CAR-T therapy in AML and expands the technical pathway for target screening [61]. Potential target antigens for CAR-T therapy in AML are detailed in Table 1.

In addition to the above targets, a series of representative preclinical and clinical studies have further validated the translational value of CAR-T therapy for AML. Mai et al. confirmed that LILRB3 modulates AML progression and serves as a potent target for CAR-T cell therapy, with LILRB3-targeted CAR-T cells exerting robust antitumour effects *in vivo* [74]. For the classical target CD33, Tambaro et al. conducted a phase I clinical trial of autologous CD33-CAR-T cells in relapsed/refractory AML, demonstrating the feasibility of CAR-T production and manageable toxicities despite limited clinical responses at the lowest dose level [75]. CYAD-01, an NKG2D-based CAR-T product, exhibited favourable safety and preliminary antileukemic activity in a phase I trial for relapsed/refractory AML and myelodysplastic syndromes, supporting the concept of targeting NKG2D ligands [76]. CD123 has been extensively explored as a promising target: Mardiros et al. first verified the specific cytolytic and antitumour effects of CD123-CAR-T cells against human AML [77], while switchable universal UniCAR-T-CD123 platforms were developed to improve safety, with preclinical proof of concept [78] and clinical proof of concept in relapsed/refractory AML patients [79]. C-type lectin-like molecule 1 (CLL-1) is another ideal target absent on normal haematopoietic stem cells, and CLL-1-CAR-T cells showed effective anti-leukaemia activity in preclinical models [80]. Moreover, mutated GM-CSF-based CAR-T cells targeting CD116/CD131 complexes exhibited enhanced anti-tumour effects against AML [81], and anti-CLL1 CAR-T therapy achieved promising efficacy in paediatric [82,83] and adult [84] relapsed/refractory AML patients in clinical trials. TIM3 has been identified and validated as a novel target antigen for CAR-T cells in AML [85]. TIM3-targeted CAR-T cells demonstrate specific and potent antileukemic activity against AML cells while sparing normal haematopoietic stem and progenitor cells, supporting its promise as a safe and effective target for AML immunotherapy.

Table 1. Potential target antigens for CAR-T therapy in AML.

Target	Disease types	<i>In Vivo/In Vitro</i> /Clinical Trial	Expression characteristics	Mechanisms	Challenges	References
CLL-1	EMDs in patients diagnosed with AML or AML	Clinical Trial	specific markers of AML stem cells	regulate the release levels of cytokines (CRP, IL-2, IL-4, IL-6, IL-10, TNF- α , IFN- γ and ferritin)	This study was conducted at a single centre and included only 47 patients. In future research, expanding the sample size is necessary to validate these results again.	[62]
CD37	AML	<i>In Vitro</i> (HEK-P, BL-41, GRANTA-519, Daudi, K-562, MOLM-13, SKM-1, HL-60, U-937, MV4-11, HEL, Jurkat76)/Clinical Trial/ <i>In Vitro</i> (NXG mice)	The mature B-cell marker CD37 is expressed in AML samples, and its presence is associated with the European LeukemiaNet 2017 risk stratification.	It specifically kills AML cells, secretes proinflammatory cytokines (TNF- α , IFN- γ , GM-CSF, G-CSF, IL-2, IL-12, IL-15, IL-17 and MIP-1 β), and controls cancer progression <i>in vivo</i> , without toxicity to haematopoietic stem cells.	focused on the presented 4-1BB-containing CD37CAR design; however, alternative signalling tails could be compared to identify the most efficient anti-AML CD37CAR.	[63]
CD19	relapsed CD19-positive t(8;21) AML	Clinical Trial (phase II)	A B-cell marker aberrantly expressed in AML with t(8;21).	reduction in RUNX1T1 fusion transcript levels	haematological toxicity, CRS	[64]
CD19	relapsed/refractory AML	Clinical Trial (phase II)	A B-cell marker that is aberrantly expressed in AML, predominantly associated with t(8;21)(q22.1;q22.1).		The response is transient, and allogeneic haematopoietic stem cell transplantation should be performed subsequently.	[65]
LILRB1	haematological malignancies	<i>In Vivo</i> (NSG mice)	an immune inhibitory receptor, is normally expressed only on monocytes and B-cells; stable LILRB1 expression in B-ALL and B-cell non-Hodgkin lymphoma, even after CD20/CD19-based immunotherapies	It provides significant advantages by preserving HSCs and neutrophils, potentially avoiding severe immunosuppression, and enabling the elimination of monocytic and mixed-lineage AML.	focus on refining the scFv and exploring its integration into alternative CAR backbones to enhance therapeutic efficacy	[66]
CG1/A2	HLA-A2+ AML and CML	<i>In Vitro</i> (U937 AML, T2 B/T lymphoblastic cells, and 293/17 human embryonic kidney cell lines as well as Patient samples and healthy donor peripheral blood) and <i>In Vivo</i> (H2L2 human transgenic mice)	HLA-A2-restricted cathepsin G-derived signal peptide CG1	Effective killing ability without affecting normal bone marrow haematopoiesis.	lack of clinical application	[67]
5T4	AML	<i>In Vitro</i> (Human AML cell lines Kasumi-1, THP-1 and AML-193) and <i>In Vivo</i> (NCG mice/THP-1 xenograft model)	Approximately 50% of AML patients exhibit 5T4 expression.	Specifically kills AML tumour cell lines, including LSCs, with safety.	A promising target for immunological intervention in AML.	[68]
CD155	AML, NSCLC and pancreatic cancer	<i>In Vivo</i> (NSG-SGM3 mice and C57BL/6J mice) and <i>In Vivo</i> (Human AML cell lines (wild-type and CD155-KO MOLM13, U937 and THP-1), ALL cell line NALM6, and murine AML cell lines (wild-type and CD155-KO C1498))	An adhesion molecule upregulated during tumour progression, serving as a target for CAR-T cell therapy in leukaemia.	potent antitumour activity	CD155 can be safely and effectively targeted by CD155 CAR-T cells, providing a new strategy for the treatment of AML. However, the clinical translation of allogeneic and off-the-shelf universal CAR-T cell therapies still faces numerous challenges, and whether they can truly reduce the cost of cellular immunotherapy remains to be verified.	[69]

(Continued)

Table 1. Continued.

Target	Disease types	In Vivo/In Vitro/Clinical Trial	Expression characteristics	Mechanisms	Challenges	References
CD70	AML	<i>In Vitro</i> (HEK293T/17, K562, HL-60, Kasumi-1, U937, KG-1a, THP-1, MOLM-13)/ <i>In Vivo</i> (NSG mice)	It is heterogeneously expressed in primary AML cells (including leukemic blasts, progenitors and stem cells), while normal haematopoietic stem cells and most blood cells do not express it.	It exhibits strong cytotoxicity, cytokine production and proliferative capacity against CD70+ AML cell lines. In the Molm-13 xenograft mouse model, it also demonstrates antileukemic activity and prolongs survival time, but does not completely eliminate leukaemia <i>in vivo</i> .	This therapy needs to be optimized by innovatively combining CAR structures or increasing the expression density of CD70 on the surface of leukemic cells.	[70]
Flt3m	AML	<i>In Vitro</i> (Human HEK293T, Jurkat, THP-1 and U937 cells)	Flt3m-CAR-T cells with a mutant Flt3 ligand (Flt3Lg-L27P) as the recognition domain	Compared with wild-type Flt3Lg, the mutant Flt3Lg-L27P exhibits a more than 10-fold reduction in biological activity, which reduces cyclic downregulation of Flt3 receptors and activation of proliferative signals in leukaemia cells, while preserving targeting specificity for Flt3-positive cells. Flt3m-CAR-T cells combine high specificity of ligand-receptor recognition with a lower risk of promoting proliferation.	Provides a new strategy for safety optimization of CAR-T therapy in AML.	[71]
CD38	MM, T-ALL and AML	<i>In Vitro</i> (MOLM14 cell line, primary patient specimens of MM, T-ALL and AML, Primary human CD34+ cells)/ <i>In Vivo</i> (NSG mice transgenic for human IL-3, SCF and granulocyte-macrop age colony-stimulating factor)	It is expressed in MM and T-ALL, and also overexpressed in AML.	Effectively rejects tumour cells and prolongs survival time, but may lead to a decrease in normal haematopoietic progenitor cells. Potential toxicity needs to be monitored during clinical translation.	The therapeutic potential for multiple CD38-expressing malignancies expands the possibility of new therapies impacting different patient populations.	[72]
CD41	AMKL, a rare subtype of AML commonly observed in children with Down syndrome	<i>In Vitro</i> (DAMI Luc2)	Specific surface antigens of M7-AMKL	93.8% of CAR-T cells were eGFP-positive. Although the acute effect was limited, co-culture for 24 h at a low effector-to-target (E: T) ratio disrupted the membrane integrity of target cells and reduced the number of M7-AMKL cells, while the cytotoxicity in the high E: T ratio group was not significant.	Anti-CD41 CAR-T cells exhibit cytotoxicity against AMKL cells at low E:T ratios and within a limited time frame, providing preclinical evidence for CAR-T therapy in this leukaemia subtype.	[73]

3.2. Target optimization strategies

3.2.1. Reducing antigen escape

3.2.1.1. Single-target CAR-T. In CAR-T cell therapy for AML, single-target optimization strategies targeting the classical antigen CD33 have achieved breakthrough progress through precise regulation of antigen epitopes and pharmacological intervention. The dimerizer-adjusted immune receptor complex (DARIC33) system developed by Appelbaum et al. as a rapamycin-regulated CD33-targeting CAR-T platform, enables reversible activation of T cell effector functions *via* nanomolar (1 nM) rapamycin [86]. This therapy retains high-specificity cytotoxicity against leukemic cells while avoiding inhibition of colony-forming capacity in CD34+ haematopoietic stem cells, establishing an innovative paradigm for pharmacologically precise regulation of CAR-T cells.

Aiming at the limitation that traditional CD33-targeted therapies mostly recognize the distal membrane v-set domain, Fiorenza's team developed CD33PAN CAR-T cells targeting the proximal membrane c2-set domain. This strategy significantly enhances the killing efficiency against AML cells by shortening the action distance between T cells and tumour cell membranes [87]. Preclinical studies show that CD33PAN CAR-T cells not only effectively eliminate tumours in tumour-bearing mice and improve survival rates but also reduce the risk of immune escape in CD33 low-expression or negative cells. Similarly, 3P14HLh28Z CAR-T cells developed by immunizing humanized mice with the proximal membrane fragment of CD33, using high-affinity binders to target proximal membrane epitopes, exhibit better tumour control and survival benefits than lintuzumab-derived CAR-T cells targeting distal epitopes in low-antigen density environments and patient-derived xenograft models [88].

Collectively, these studies confirm that optimizing antigen epitope recognition sites (e.g. proximal membrane targeting) and introducing controllable regulatory mechanisms can effectively enhance the resistance of single-target CAR-T cells to antigen escape. Additionally, addressing on-target/off-tumour toxicity in CAR-T therapy, research has used base editors (BEs) and prime editors (PEs) to modify CD123 epitopes on haematopoietic stem and progenitor cells [89]. This approach achieves selective resistance of bone marrow lineages to CAR-T immunotherapy in humanized mouse models, offering a new strategy for relapsed AML. Exploiting PRAME's high expression in AML and lack of expression in normal haematopoietic cells, PRAME mTCRCAR-T cells were developed using the TCR-mimic antibody Pr20 to recognize PRAME peptide-HLA-A*02 complexes. These cells showed potent target-specific anti-leukemic efficacy, with IFN- γ treatment further enhancing their function, and achieved significant leukaemia control *in vivo* [90].

For FLT3-mutated AML, the development of monovalent FLT3-CART and bispecific CD19 $\times$ FLT3-CART further expands the single-target optimization strategy [91]. FLT3-CART demonstrates significant cytotoxicity and cytokine production against FLT3-mutated AML cell lines, with extremely low cross-reactivity to normal tissues. In xenograft models, it effectively inhibits leukaemia proliferation, including in patient-derived models with lineage switching. Bispecific CAR-T cells, by co-targeting CD19 and FLT3, may further reduce the risk of antigen escape. These studies provide theoretical and technical support for precision immunotherapy of AML, from mechanism of action to clinical translation.

In the landscape of AML-targeted CAR-T development, CD123, CLL-1 and CD33 represent three core myeloid-associated antigens with extensive preclinical and clinical validation for overcoming antigen escape. CD123-targeted CAR-T therapy has achieved iterative progress from basic preclinical research to clinical transformation. Mardiros et al. laid the foundation for CD123-CAR-T development by verifying its specific killing effect on AML cells [77], and the rapidly switchable UniCAR-T-CD123 platform was subsequently constructed to solve the problem of on-target/off-tumour toxicity, with preclinical [78] and initial clinical [79] data confirming its controllable safety and encouraging efficacy. For CLL-1, preclinical studies confirmed its specificity to AML cells [80], and clinical trials further validated its therapeutic value: Zhang et al. reported that anti-CLL1 CAR-T cells achieved favourable remission rates in paediatric relapsed/refractory AML [82,83], while Jin et al. completed the first-in-human phase I study of CLL-1 CAR-T cells in adult patients, with a 70% CR/CRi rate and manageable safety profile [84]. For CD33, Tambaro et al. revealed the challenges of autologous CD33-CAR-T production and clinical application in relapsed/refractory AML, providing important references for subsequent target optimization [75].

3.2.1.2. Dual-target CAR-T. In the field of CAR-T cell therapy for AML, antigenic heterogeneity and immune escape are core challenges restricting clinical efficacy. Dual-target CAR-T technology provides an important

solution to break through this dilemma by expanding the antigen recognition spectrum and enhancing the targeting specificity of tumour cells, and multiple innovative strategies have shown significant clinical translation potential. Aiming at treatment resistance caused by antigen escape in AML, novel dual-target CAR-T cells (such as 123NL CAR-T) significantly enhance the ability to eliminate AML cells by simultaneously targeting CD123 and NKG2D ligands (NKG2DL) [92]. This design integrates the RQR8 suicide gene system into the CAR structure, which can not only selectively eliminate immunosuppressive cells (such as M-MDSCs and M2-type macrophages), but also precisely regulate CAR-T activity through rituximab-induced RQR8 activation [93]. This strategy effectively overcomes the limitations of single-target escape and immunosuppressive microenvironment, providing new ideas for the optimization of CAR-T therapy for AML.

At the mechanism level, dual-target CAR-T effectively reduces the risk of tumour escape caused by single antigen loss through the collaborative recognition of two or more antigens. For example, tandem CAR-T cells targeting HER2 and IL13R α 2 significantly alleviate antigen escape of tumour cells by acting on two tumour-associated antigens simultaneously [94]. Based on this, aiming at the heterogeneous characteristics of AML LSCs, the ADGRE2/CLEC12A dual-target CAR-T strategy achieves effective blockade of antigen escape and complete tumour eradication in models by co-targeting ADGRE2⁺CLEC12A⁺ phenotypic LSCs, while maintaining safety for normal haematopoietic stem cells [95].

In the dimension of technological innovation, the CAR-TCR-T dual-targeted cell therapy developed by Teppert et al. significantly enhances the killing activity against AML cells with low-antigen expression by co-expressing CD33-CAR and dNPM1-TCR in the same T cells. Transcriptomic analysis shows that this strategy can significantly improve the efficiency of immune synapse formation, activate key T cell signalling pathways, and enhance cell activation levels. *In vivo* experiments further confirm that compared with the simple mixture of CAR-T and TCR-T cell preparations, CAR-TCR-T dual-targeted cells can achieve complete clearance of AML xenografts, verifying the unique advantage of the dual-target co-expression strategy in overcoming antigenic heterogeneity [96].

Based on the analysis of antigen expression profiles in paediatric primary AML using single-cell sequencing technology, CD33, CD38, CD371, IL1RAP and CD123 were identified as highly expressed antigens. Aiming at the significant inter-tumoural and intra-tumoural heterogeneity of AML, the research team utilized the modular AdCAR platform. By combining AdCAR-T cells targeting the above antigens, they achieved effective control of heterogeneous AML in patient-derived xenograft models, providing a brand-new technical approach to overcome the challenge of antigen escape [97].

In clinical translation-oriented research, multiple dual-target strategies have demonstrated excellent therapeutic performance. The Loop33 $\times$ 123 dual-target CAR-T, by simultaneously targeting CD33 and CD123, not only significantly enhances T-cell degranulation efficiency and cytokine secretion levels but also exhibits stronger survival benefits than single-agent CAR-T in immune escape models, with no obvious toxicity observed [98]. Anti-IL10R CAR-T cells effectively eliminate AML cells with different antigen expression levels by secreting bispecific antibodies targeting CD33, addressing the immune escape problem caused by heterogeneous expression of IL10R and CD33 [99]. Furthermore, CD70 CAR-T cells (7033 CAR-T) secrete anti-CD33/anti-CD3 dual-target antibodies, maintaining killing activity against single-antigen knockout tumour cells, achieving complete tumour eradication *in vivo*, and significantly outperforming single-target therapies in terms of cell persistence and signalling efficiency. They can also redirect patients' endogenous T cells to participate in tumour killing [100].

In the engineering modification of antigen recognition domains, researchers constructed a bispecific antigen recognition domain (78.123) by fusing a GRP78-specific peptide segment with a CD123 scFv [101]. After linker optimization, 78.123 bispecific CAR-T cells can precisely recognize and kill AML cells expressing GRP78 alone, CD123 alone, or both antigens simultaneously. Functional validation shows that these cells exhibit stronger tumour suppressive efficacy than monospecific CAR-T cells, and characteristics such as the length and flexibility of the linker significantly affect their antigen-binding and signal transduction efficiency, providing a theoretical basis for CAR structure optimization.

Another strategy focuses on myeloid-specific antigen co-targeting. The double-chain 123CL CAR-T cells targeting CD123 and CLL1 have demonstrated significant anti-leukaemia activity in xenograft models [102]. To improve treatment safety, these cells incorporate the RQR8 marker and suicide gene system to enable controlled elimination of CAR-T cells when needed, with no evident on-target toxicity against

normal haematopoietic stem cells. Tandem bispecific CD123/CLL-1 CAR-T cells activate T-cell signalling pathways through dual-antigen collaboration, effectively overcoming antigen escape caused by AML tumour heterogeneity, and providing a new technical approach to enhance CAR-T therapy specificity and anti-tumour durability [103].

It is noteworthy that the study of engineered tandem CD33×CD146 CAR-Clk cells not only validates the feasibility of simultaneous targeting of leukemic cells (CD33) and bone marrow stromal cells (CD146), but also reveals the inhibitory effect of bone marrow stromal cells on the effector functions of CAR-Clk cells. This suggests that dual targeting strategies for both tumour cells and their microenvironment should be emphasized in the development of CAR-T therapies for AML [104].

In summary, dual-target CAR-T technology has demonstrated unique advantages in overcoming AML antigenic heterogeneity and reducing the risk of immune escape through diversified antigen combination strategies and molecular engineering optimization, providing important theoretical basis and technical support for precision immunotherapy of AML.

3.2.2. Nanobody-based targeted CAR-T therapy

In the strategy for optimizing targets in CAR-T cell therapy for AML, nanobody-based targeting technology has emerged as an innovative direction to break through the bottlenecks of traditional CAR-T therapy, leveraging its advantages of high affinity, low immunogenicity and structural stability. Multiple studies have achieved remarkable progress in enhancing antitumour activity, reducing off-target toxicity and optimizing cellular function regulation through nanobody engineering modifications.

Aiming at the toxicity of bispecific split CAR-T to normal HSCs, the third-generation homologous nanobody-targeted CD13/TIM3 bispecific CAR-T cells have emerged [105]. This design specifically recognizes CD13 and TIM3 antigens through nanobodies, exhibiting potent anti-leukemic efficacy in preclinical models, while showing negligible toxicity to human bone marrow progenitor cells and humanized mouse HSCs. This strategy significantly reduces the risk of normal tissue damage while maintaining therapeutic efficacy, providing a safe and effective new solution for clinical translation in aggressive/refractory AML.

NS7 CAR-T cells constructed based on the variable dVHH have demonstrated unique advantages in CD7 targeting [106]. These cells not only exhibit excellent CD7 binding specificity with affinity comparable to that of traditional scFv, but also significantly enhance T-cell proliferation capacity. Phase I clinical trial data show that among 10 patients with CD7+ refractory/relapsed AML, 7 achieved complete remission, and the treatment was well-tolerated. Although resistance in some non-responding or relapsed patients was caused by CD7 antigen loss, the favourable clinical response still confirms the application potential of dVHH nanobodies in CAR-T therapy.

To overcome the issues of strong signal rigidity and vulnerable binding epitopes in traditional scFv fragments, Bernasconi-Bisio et al. developed a SdAb-driven CAR-T cell screening platform [107]. Targeting the key AML antigen CD33, this platform screened various SdAbs with different affinities to construct second-generation CAR-T cells. Studies have shown that SdAb-based CAR-T cells can effectively regulate the intensity of costimulatory signals and enhance cytokine secretion capacity. In an AML xenograft mouse model, low-dose SdAb-CAR-T cells exhibited significantly better therapeutic effects than traditional scFv-based CD33 CAR-T cells, fully demonstrating the structural advantages of nanobodies in small size, high stability and low immunogenicity.

Additionally, nanobody-based CD38-CAR-T cells have also shown promising prospects in AML treatment [108]. These cells can efficiently lyse CD38+ AML cell lines (such as NB4 and U937) and primary tumour cells at an effector-to-target ratio as low as 1:16. By downregulating PI3K δ expression *via* lentiviral vectors, the further optimized CD38-CAR-T cells maintain anti-leukaemia activity while significantly reducing excessive release of cytokines (e.g. IL-2 and IFN- γ), thereby minimizing the risk of CRS. In an AML mouse model, this optimization strategy significantly improved survival compared with the unmodified group, confirming the critical role of nanobody targeting combined with PI3K δ regulation in balancing therapeutic efficacy and safety.

In conclusion, nanobody-based CAR-T technology provides an innovative and safe target optimization strategy for AML treatment through precise antigen recognition, signalling pathway optimization and functional regulation, holding promise to become an important technical support for promoting the clinical translation of CAR-T therapy.

3.2.3. Logic-gated CARs

Key progress has been made in logic-gated CAR technology for target optimization in AML CAR-T therapy. Traditional CAR-T therapies suffer from inherent limitations such as high off-target toxicity and insufficient antigen recognition specificity. In contrast, CD33/CD123 bispecific CAR-T cells based on 'IF-THEN' synchronous Notch gating design have effectively addressed these drawbacks by innovatively establishing a dual-antigen collaborative recognition mechanism [109]. This technical approach enhances therapeutic performance through multiple mechanisms: First, the dual-antigen co-recognition mode significantly reduces off-target toxicity; second, it efficiently induces T-cell differentiation towards a memory phenotype and enhances *ex vivo* expansion capacity; third, it decreases the incidence of severe adverse reactions such as CRS. This strategy substantially improves tumour-killing specificity while effectively protecting normal HSPCs, successfully achieving a dynamic balance between treatment safety and efficacy.

Another study on bispecific CD33/CD123 CAR-T cells based on 'AND' logic gate design employs a unique signal domain separation technique, integrating the 4-1BB costimulatory domain and CD3 ζ signalling domain into distinct scFv structures, respectively [110]. This design enables CAR-T cells to activate only when recognizing both antigens simultaneously, thus achieving precise regulation of cytokine secretion and cytotoxicity. *In vivo* animal model studies confirm that it preserves normal tissue function while demonstrating therapeutic efficacy comparable to monospecific CAR-T, providing an innovative technical solution for target optimization in AML.

Aiming at the common challenges of CAR-T cell 'off-target toxicity' and relapse caused by antigen escape in the treatment of haematological malignancies such as AML, the multi-target non-gated CAR-T technology (Tmod™ system) has systematically optimized target selection through an innovative logic gating strategy [111]. This technology constructs dual-functional signalling pathways: at the tumour targeting recognition level, CD33 (an antigen co-expressed by tumour cells and normal bone marrow cells) and CD43 (an antigen widely expressed in the haematopoietic system) are selected as targets to design single/bispecific activating receptors, enhancing tumour cell recognition efficiency; at the normal tissue protection level, antigens specific to normal blood cells (lowly expressed in tumour cells) such as CD16b and CLEC9A are targeted to construct single/bispecific inhibitory receptors, precisely avoiding normal tissue damage. By integrating dual signals of activation and inhibition, the Tmod™ system achieves an intelligent 'logic gating' effect, which not only significantly improves the selectivity of tumour killing but also breaks through the technical bottleneck of traditional CAR-T therapy. This innovative mechanism provides a new strategy for AML treatment that balances efficacy and safety, and is expected to lead the innovation of AML treatment models.

3.2.4. Target optimization by adapter-CAR technology

In the target optimization system of CAR-T therapy for AML, adapter-CAR technology has opened up a new technical pathway through an innovative molecular bridging mechanism. Aiming at the core defects of traditional CAR-T therapy, such as tumour antigen escape and strong dependence on single targets, adapter-CAR technology uses bispecific antibodies to construct functional modules. One end of the module targets the commonly used GGGGS linker on the surface of CAR-T cells, and the other end specifically recognizes tumour antigens, thus achieving efficient bridging between CAR-T cells and tumour cells [112]. Mechanistic studies have shown that this technology can significantly enhance the activation efficiency, *ex vivo* proliferation capacity and tumour cell killing activity of CAR-T cells. By flexibly regulating the antigen-binding specificity of BsAbs, dose-dependent redirectional recognition of different tumour antigens by CAR-T cells can be achieved. The above characteristics make adapter-CAR technology show significant advantages in overcoming relapsed/refractory AML, reducing treatment-related toxicity and improving therapeutic efficacy.

In addressing the safety and antigen escape issues of CAR-T cell therapy for myeloid malignancies, FLT3-directed adapter CAR-T cells developed by Peschke's team have demonstrated outstanding performance [113]. *In vivo* mouse models validated their therapeutic efficacy, with PET analysis demonstrating that the adapter module has a short serum half-life, endowing the system with rapid activation and shutdown properties, and thus providing robust preclinical evidence for clinical translation.

To address the technical bottlenecks of CAR-T cells in AML treatment, including single antigen targeting, narrow therapeutic window and lack of dynamic regulation, the AdFITC-CAR-T cell platform achieves precise matching to AML antigen expression profiles and conditional activity regulation through flexible configuration of aptamer modules [114]. In therapeutic xenograft mouse models, AdFITC-CAR-T cells combined with a single aptamer exhibit comparable efficacy to traditional direct-targeted CAR-T, while combined aptamer application further improves the therapeutic efficacy against cell lines and primary AML, providing theoretical and practical evidence for optimizing AML immunotargeted therapy based on aptamer combination strategies.

When optimizing the CD4-targeted CAR-T cell therapy system, Abdulla's team developed a microfluidic chip technology based on the CD8 aptamer A3t-MU, achieving efficient capture and non-destructive release of CD8⁺ T cells [115]. CD8⁺ T cells purified using this chip and successfully transduced with CD4-CAR demonstrated significantly enhanced targeting activity in *in vivo* antitumour models, providing a low-cost, high-efficiency aptamer-chip separation strategy for generating high-purity, high-activity CD4-CAR-T cells.

Aiming at the common challenges of target heterogeneity and T-cell exhaustion in AML treatment, the Fab fragment-based AdCAR platform technology achieves sequential multi-targeted regulation by dynamically switching adapter molecules against CD33, CD123, CLL-1, etc [116]. Studies in primary tumour models have confirmed that this strategy can not only effectively overcome antigen escape, but also intermittent AdCAR stimulation significantly alleviates exhaustion caused by chronic T-cell activation, maintaining long-term antitumour activity. This provides an innovative solution to break through the drug resistance dilemma in AML immunotherapy.

3.2.5. Optimized combination therapy

In the field of CAR-T therapy for AML, optimizing the timing of combination therapy has emerged as a core strategy to enhance clinical efficacy and reduce adverse reactions. From the perspective of precise control of clinical intervention windows, prophylactic use of inhibitors before CAR-T cell infusion or timely intervention during the CRS phase can effectively manage adverse reactions while avoiding interference with CAR-T cell expansion and maintaining their durable antitumour activity. Clinical practice has demonstrated significant advantages of combination therapy strategies [117]. In patients with relapsed/refractory AML, consolidation with allogeneic HSCT following CAR-T-induced remission has been explored to improve disease control. For instance, a 73-year-old patient with refractory AML achieved morphological complete remission after CLL1-targeted CAR-T therapy, and subsequent consolidation with allo-HSCT led to sustained MRD-negative remission [118]. Donor-derived CAR-T therapy followed by timely allo-HSCT consolidation has also been shown to improve long-term outcomes in this population. Further optimization of CAR-T dosing, sequential infusion schedules, and CD4⁺/CD8⁺ subset ratios can enhance *in vivo* persistence, reduce toxicity, and improve antitumour efficacy, particularly for patients relapsing after allo-HSCT [119].

At the level of drug combination therapy, demethylating agents (such as decitabine and azacitidine) exhibit multiple synergistic mechanisms [120]. On one hand, they optimize CAR-T cell function by increasing the number and persistence of CAR⁺ cells and inducing T-cell differentiation towards activated and memory phenotypes (with decitabine showing more significant effects). On the other hand, they promote AML cell apoptosis, particularly in DNMT3A-mutated tumour cells, and enhance tumour recognition and killing by CAR-T cells through upregulating CD44v6 expression in tumour cells. *In vivo* studies confirm that CAR-T cells combined with demethylating agents yield potent antitumour activity, indicating that demethylating therapy combined with CD44v6 CAR-T cells represents an effective strategy for optimizing AML treatment. Additionally, the abnormal tumour microenvironment is a key factor leading to T-cell exhaustion in AML. AML cells construct complex drug resistance mechanisms by secreting cytokines such as GM-CSF, FLT3L and IL-3, significantly inhibiting the killing function and *in vivo* persistence of autologous anti-CD123 CAR-T cells. Aiming at this pathological mechanism, the combined application of JAK inhibitors (such as ruxolitinib) or BCL-2 inhibitors (such as venetoclax) with CAR-T therapy is expected to break through the clinical bottleneck of insufficient persistence of CAR-T cells in AML treatment, providing an innovative therapeutic approach to overcome current efficacy limitations and improve patient prognosis [121].

4. Impact of AML immunomicroenvironment on CAR-T function

Inhibitory factors in the tumour immunomicroenvironment (such as high expression of immune checkpoints, cytokine imbalance and enrichment of myeloid-derived suppressor cells) significantly restrict the efficacy of CAR-T cells in AML. AML immune microenvironment is characterized by prominent dynamic evolution and spatial heterogeneity, which further shape the regional immunosuppressive gradients, clonal immune escape and CAR-T cell dysfunction in bone marrow niches. Leukemic clones at different disease stages or anatomical sites (bone marrow, peripheral blood, extramedullary lesions) exhibit distinct immune profiles, accompanied by stage-dependent remodelling of extracellular matrix, nutrient competition and metabolite accumulation. Such spatiotemporal heterogeneity not only reduces the intratumoural infiltration and durable activity of CAR-T cells but also drives regional immune tolerance and early relapse. Currently, remodelling the AML immunomicroenvironment through multidimensional intervention strategies has become a key direction for enhancing CAR-T function, specifically including:

4.1. Combination with immune checkpoint inhibitors

In the immunotherapy research for AML, CAR-T therapy targeting LILRB3 has demonstrated a unique dual regulatory mechanism against tumours. As an inhibitory receptor on myeloid cell surfaces, LILRB3 exhibits inducible expression characteristics under inflammatory stress or chemotherapy intervention. This key finding was systematically validated by Mai et al. who demonstrated that blockade of LILRB3 inhibits leukaemia cell proliferation and progression, while LILRB3-targeted CAR-T cells exhibit potent preclinical antitumour efficacy. By blocking the LILRB3 signalling pathway, it can activate the cascade reactions of key transcription factors for myeloid differentiation (such as PU.1, C/EBP family, IRF), while effectively inhibiting the cell proliferation signalling network mediated by AKT, cyclin D1 and phosphorylated Rb protein, thus achieving effective blockade of leukaemia progression. In contrast, LILRB3 agonists can provide survival advantages for leukaemia cells by activating the cholesterol metabolism pathway. The discovery of this target and the intervention strategy have opened up a new research direction for immune checkpoint therapy in AML [74].

The combined application of CAR-T cell therapy and immune checkpoint inhibitors, through multidimensional regulation of the tumour microenvironment, has become an important strategy to overcome T-cell exhaustion. Clinical research data show that this combination therapy regimen exhibits significantly better clinical efficacy than monotherapy in AML treatment, particularly with potential advantages in mitigating CAR-T cell off-target effects and controlling cytokine release syndrome. However, the risk of immune-related adverse events induced by combination therapy requires careful attention. It is recommended to establish individualized treatment protocols to achieve dynamic balance between efficacy and safety. Additionally, a prognostic model constructed based on BATF and EGR1 gene expression levels can effectively distinguish survival differences among patients. A risk scoring system integrating immune checkpoint genes such as PD-1, PD-L1 and TIM-3 can accurately predict the overall survival of AML patients and allo-HSCT recipients, offering innovative evidence for disease prognosis assessment and treatment decision-making [122].

The synergistic mechanisms of combined CAR-T cell and immune checkpoint inhibitor therapy in AML can be summarized at the following three levels [123,124]: First, at the level of target complementarity, CAR-T cells achieve specific tumour antigen recognition and direct cytotoxicity through their CARs, while immune checkpoint inhibitors (e.g. anti-PD-1, anti-CTLA-4 antibodies) block immunosuppressive signalling pathways such as PD-L1/PD-1 and B7/CTLA-4, thereby reversing tumour-mediated T-cell suppression. This combination generates a synergistic effect of targeted killing and immune activation. Second, in terms of reversing T-cell exhaustion, persistent antigen stimulation, immunosuppressive metabolites (e.g. adenosine, lactate) and inhibitory cell populations (e.g. Tregs, MDSCs) in the tumour microenvironment collectively drive T cells into an exhausted state (characterized by high expression of inhibitory receptors such as PD-1 and TIM-3). The combined therapy counteracts this by using immune checkpoint inhibitors to block inhibitory signals while CAR-T cells reduce antigen load through tumour cell clearance, thereby synergistically restoring T-cell cytotoxic function. Finally, in remodelling the tumour microenvironment, immune checkpoint inhibitors modulate the cytokine network within the tumour microenvironment,

suppressing immunosuppressive cytokines (e.g. IL-10, TGF- β) while promoting Th1-type cytokine expression. This enhances CAR-T cell homing, persistence and antitumour efficacy, establishing a positive feedback loop of 'tumour killing - tumour microenvironment normalization - immune potentiation', ultimately maximizing synergistic antitumour effects.

4.2. Gene editing technologies

In the field of CAR-T therapy for AML, gene editing technologies have emerged as a core strategy to optimize CAR-T cell function by targeting key genes. Studies on CRISPR-Cas9-based gene editing show that CD97-knockout CAR-T cells, generated by knocking out the CD97 gene in CAR-T cells, efficiently eliminate AML cell lines and primary tumour cells while exhibiting low cytotoxicity towards HSPCs [125]. Further optimization of the signalling pathway through modification of the CD3 ζ domain enables the improved CD97 CAR-T cells to demonstrate durable antitumour activity in xenograft models. This mechanism is closely associated with delaying cell differentiation and inhibiting exhaustion to maintain functional stability. Additionally, addressing the 'self-killing' issue caused by high CD45 expression in T cells and NK cells when using CD45 as a target, CRISPR/Cas9-mediated CD45 gene knockout has successfully generated CD45ko/CD45CAR-T cells. These cells not only effectively avoid autophagic killing but also maintain potent and sustained antitumour activity against CD45-expressing AML and lymphoma cell lines, without compromising the proliferative capacity of T/NK cells or the functional integrity of CAR-T cells. This provides a novel optimization direction for CAR-T therapy in AML, holding potential application value in preconditioning regimens for stem cell transplantation [126].

In-depth studies have revealed that abnormally high expression of methyltransferase *METTL16* in AML mediates m6A hypermethylation modification of TCF-1, leading to decreased mRNA expression levels and stability. This induces T-cell exhaustion, impairs self-renewal capacity and ultimately suppresses the persistence and tumour-killing function of CAR-T cells [127]. Experiments have confirmed that knockout of the *METTL16* gene promotes differentiation of CAR-T cells into pre-exhausted progenitor T cells, effectively enhancing cell persistence, memory phenotype formation and antitumour activity. These findings indicate that gene editing strategies targeting *METTL16* hold promise to improve CAR-T cell function by regulating m6A methylation modification, opening a new avenue for immunotherapy of AML.

Aiming at the functional defects of T cells in AML patients caused by the tumour immunomicroenvironment (characterized by dominant effector phenotypes, high expression of exhaustion markers such as PD-1 and LAG3, and loss of memory phenotypes), the research team combined CRISPR genome editing technology with a Sleeping Beauty transposon virus-free transfer system to successfully construct allogeneic CAR-T cells with knockout of HLA-I and TCR complex (HLA-IKO/TCRKO CAR-T) [128]. This gene editing strategy primarily serves to eliminate CAR-T cell immunogenicity and abrogate alloreactivity, thereby preventing allograft rejection and graft-versus-host disease, while the one-step efficient editing platform preserves the phenotypic stability and anti-tumour function of CAR-T cells, avoiding the aggravation of T cell dysfunction associated with traditional editing methods. This technology enables one-step efficient gene editing, with edited cells maintaining consistent phenotypic characteristics and *in vivo* antitumour activity compared to unedited CAR-T cells. It also features high safety of vector integration and no obvious off-target effects. Genomic analysis further validated the high specificity of HLA-I and TCR editing, and large-scale production processes showed that these cells meet clinical application standards. The HLA-IKO/TCRKO modification addresses the core risks of allogeneic CAR-T application (rejection and GVHD) rather than directly correcting T cell functional defects in AML; meanwhile, the high-efficiency editing system retains the effector function of CAR-T cells, which indirectly supports the overcoming of autologous T cell dysfunction bottlenecks in AML patients. By remodelling the immunogenicity of CAR-T cells, this technology effectively overcomes key bottlenecks such as dysfunction of autologous T cells in AML patients and allograft rejection, providing an innovative solution to improve CAR-T cell function in the tumour immunomicroenvironment.

In optimizing the bone marrow homing capacity of CAR-T cells, researchers used gene editing technology to overexpress wild-type CXCR4 (CXCR4wt) and mutant CXCR4R334X in CD33 CAR-Clk cells, respectively [129]. *In vitro* experiments showed that overexpression of both CXCR4wt and CXCR4mut significantly enhanced the chemotactic response of CAR-Clk cells to CXCL12 secreted by bone marrow stromal cells. *In vivo* studies confirmed that CXCR4 overexpression effectively promoted the homing of

CAR-ClK cells to the bone marrow microenvironment, among which CAR-ClK cells co-expressing CXCR4wt exhibited more persistent antileukemic activity and significantly prolonged the survival of tumour-bearing mice, suggesting that CXCR4 may participate in regulating CAR-ClK cell function through non-homing-dependent mechanisms. Additionally, phase I clinical trial data showed that overexpression of the C-JUN gene significantly improves the efficacy of CAR-T cells in AML treatment, with its mechanism of action mainly involving ERK signalling pathway activation or transcriptional regulation, thereby maximizing the restoration of CAR-T cell antitumour activity, and this effect is not directly associated with anti-cell exhaustion mechanisms [130]. These studies indicate that targeted regulation of homing-related receptors and key transcription factors *via* gene editing technology provides a systematic solution for optimizing the colonization capacity and antitumour effect of CAR-T cells in the AML bone marrow microenvironment.

4.3. Cytokine armouring

In the complex regulatory system of the AML immunomicroenvironment, the impact of the tumour microenvironment on CAR-T function has become a key factor restricting therapeutic efficacy. Studies have shown that T cells and NK cells from AML patients often harbour TP53 gene mutations. T cells carrying single allelic TP53 mutations exhibit significantly reduced cytotoxicity and upregulated expression of inhibitory receptors. CAR-T cells containing clinically relevant p53 mutations (such as Y220C and R175H) show obvious exhausted phenotypes, leading to severe impairment of their antitumour functions. To address this challenge, the application of targeted small-molecule reactivators to restore the wild-type conformation of mutant p53 can effectively rescue CAR-T cell function and significantly reduce the expression levels of exhaustion markers, providing an innovative strategy to improve CAR-T cell function in the AML immunomicroenvironment [131]. Furthermore, BRD4 inhibitors can effectively block the terminal differentiation process of exhausted T cells by inhibiting BATF activity and expression, while activating and upregulating EGR1 expression, opening a new path for optimizing the clinical efficacy of CAR-T cell therapy in AML treatment [132].

In the exploration of cytokine armouring strategies, CLL-1 CAR-T cells expressing transgenic IL-15, in combination with TNF- α -blocking antibodies and activation of the caspase-9 control switch, have achieved comprehensive enhancement of cell expansion capacity, survival duration and antileukemic activity [133]. This combined intervention strategy precisely regulates the cytokine network, effectively avoiding toxicity risks caused by excessive cytokine release and abnormal accumulation of activated T cells, and successfully achieves dynamic balance between treatment safety and efficacy.

Additionally, novel ligand-designed CAR-T cells targeting IL-10R have made significant progress in AML therapy research [134]. *In vivo* animal model studies have confirmed that these engineered CAR-T cells exhibit highly efficient specific killing capacity against AML cells, while demonstrating good safety towards normal haematopoietic systems (except monocytes). Even when IL-10 is integrated into the CAR-T structure to induce phenotypic changes, their cell viability, biological activity and tumour-killing efficacy are not significantly affected. This achievement provides a brand-new target and technical pathway for CAR-T therapy in AML, holding promise to promote clinical breakthroughs in AML immunotherapy.

4.4. Multitarget tandem strategy

Aiming at the clinical challenges of high relapse rate in paediatric AML and therapeutic limitations of allo-HSCT, systematic molecular biology analysis has revealed that *KIT*, *MPL* and *FLT3* genes exhibit significantly high expression in paediatric AML samples. Based on this, the research team innovatively developed trivalent chimeric antigen receptor T cells (ELECTRIC CAR-T cells), which achieve simultaneous targeted recognition of the above three tumour-associated antigens through a single multifunctional receptor design [135]. Mouse xenograft tumour model studies show that ELECTRIC CAR-T cells exhibit potent antileukemic cytotoxic effects. This multitarget tandem strategy not only provides an innovative technical pathway for target optimization in AML CAR-T therapy but also holds promise for constructing genotoxicity-free antileukemic conditioning regimens, effectively reducing therapy-related toxicity while improving treatment efficacy. It offers new research ideas and practical directions for overcoming therapeutic challenges in paediatric AML.

4.5. Metabolic microenvironment intervention strategies

The metabolic reprogramming within the AML bone marrow tumour microenvironment serves as a pivotal obstacle undermining the therapeutic efficacy of CAR-T cell therapy, which has been widely documented in AML immunotherapy research [136]. This aberrant metabolic landscape is typified by heightened aerobic glycolysis (Warburg effect), severe glucose depletion, lactate accumulation, hypoxic conditions and dysregulated amino acid and lipid metabolism [47,137]. Such metabolic disorders not only trigger competitive nutrient deprivation that cripples CAR-T cell function and induces T cell exhaustion but also constructs a profoundly immunosuppressive milieu, thereby restricting the expansion, persistence and cytotoxicity of CAR-T cells [138,139]. Targeted modulation of the metabolic microenvironment has emerged as a viable strategy to restore the metabolic fitness and antileukemic activity of CAR-T cells, representing a key direction to surmount CAR-T therapy resistance in AML [140].

Firstly, hypoxia and the activated HIF-1 α signalling axis in the AML bone marrow microenvironment are core metabolic immunosuppressive cues. Hypoxic stress can upregulate the expression of PD-L1, glycolytic enzymes and adenosine-generating molecules in the TME, which directly attenuate the persistence and tumouricidal capacity of CAR-T cells [141]. Preclinical studies have validated that hypoxia-activated prodrugs, HIF-1 α small-molecule inhibitors, or oxygen-carrying biomaterials can effectively alleviate hypoxic immunosuppression, boost CAR-T cell infiltration into leukemic lesions, and reinforce their effector functions [142].

Secondly, the extracellular acidosis and high lactate levels stemming from the Warburg effect of AML blasts severely disrupt T cell receptor signalling transduction and pro-inflammatory cytokine secretion in CAR-T cells, exacerbating T cell dysfunction [143]. Multiple metabolic intervention strategies have been developed to reverse this suppression, including pharmacological inhibition of lactate dehydrogenase A to reduce lactate production, neutralization of extracellular acidic pH, and genetic engineering of CAR-T cells to enhance acid tolerance, all of which can mitigate metabolic inhibition and sustain CAR-T cell activity [47].

Thirdly, fierce nutrient competition – mainly for glucose, glutamine and tryptophan – between AML blasts and CAR-T cells drastically limits CAR-T cell proliferation and memory cell formation, which is a critical cause of poor therapeutic durability [144]. Emerging metabolic engineering strategies have targeted this bottleneck: overexpression of glucose transporters such as GLUT1 and GLUT5 in CAR-T cells can enhance nutrient uptake capacity, with GLUT5 particularly enabling CAR-T cells to utilize fructose abundant in the AML bone marrow microenvironment to maintain metabolic homeostasis [47,65]; additionally, metabolic gene editing and combined administration of glutaminase inhibitors can empower CAR-T cells with superior metabolic competitiveness, enabling them to compete effectively with leukemic cells for limited nutrients [47,60,145]. Precise target selection remains critical for avoiding on-target/off-tumour toxicity, and single-cell analysis has clarified the expression patterns of CAR-T target antigens in normal haematopoietic tissues and AML cells, providing a foundational reference for safe and effective CAR-T design [146].

Fourth, tumour-infiltrating CAR-T cells in the AML microenvironment have been verified to exhibit high expression of exhaustion markers and severe metabolic insufficiencies, which further restrict their effector function. Among potential targets, CD105 (endoglin) has been identified as a promising surface molecule expressed on acute leukemic cells, and CD105-targeted CAR-T therapy has shown anti-leukemic efficacy in preclinical models [147]. Genetic optimization or combined metabolic modulation can be applied to improve metabolic fitness and reduce exhaustion of CD105-directed CAR-T cells, thereby synergistically enhancing the anti-leukaemia effect [148].

Collectively, metabolic microenvironment intervention stands as a highly promising approach to overcome CAR-T cell resistance in AML [149]. The rational combination of metabolic regulation strategies and CAR-T cell engineering innovation is expected to break through multiple restrictive factors including nutrient competition, extracellular acidosis, tissue hypoxia and immunosuppressive metabolite accumulation in the AML TME, ultimately elevating the long-term persistence and antileukemic efficacy of CAR-T cell therapy [150]. Beyond this, targeting metabolic microenvironment reprogramming together with strategies to improve T-cell heterogeneity and rational CAR-T cell allocation can synergistically surmount

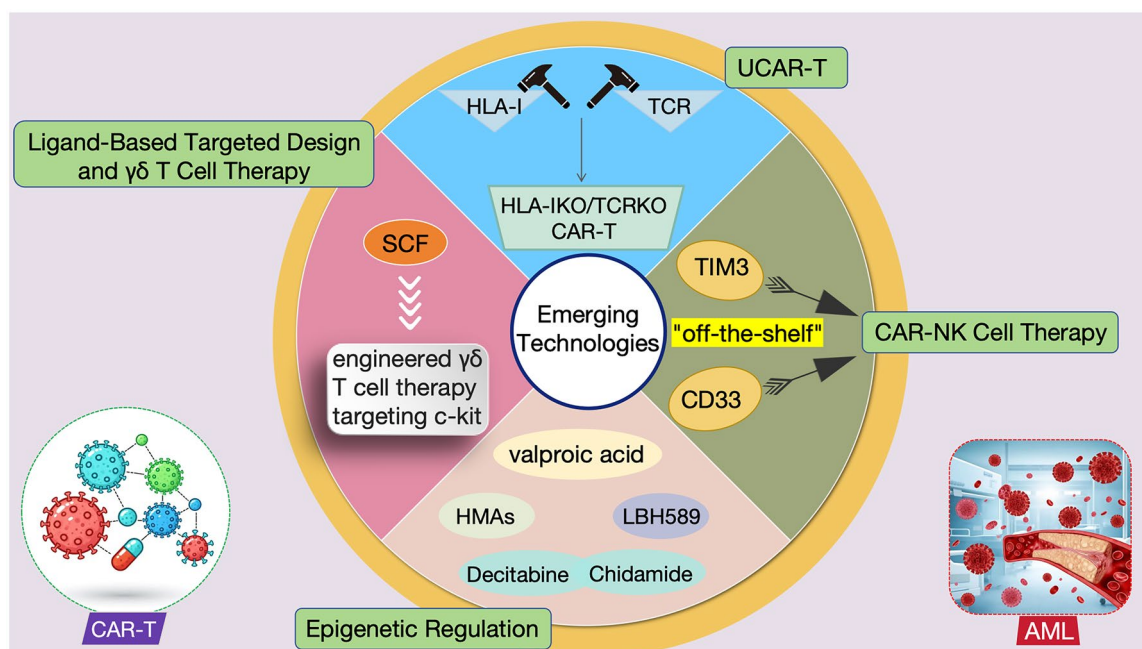


Figure 3. Emerging preclinical technologies of CAR-T therapy for AML.

the immunosuppressive and metabolic barriers in AML, further enhancing the durability and clinical efficacy of CAR-T therapy.

5. Emerging technologies in CAR-T therapy

The 2024 American Society of Haematology annual meeting witnessed significant breakthroughs in the field of AML-targeted CAR-T therapy [151]. A series of innovative technologies have emerged and demonstrated enormous potential, including UCAR-T technology, CAR-NK therapy, epigenetic regulation strategies, ligand-based targeted design, and $\gamma\delta$ T cell therapy, among other cutting-edge directions (Figure 3).

5.1. UCAR-T

Although CLL1-targeted CAR-T cell therapy has provided a new therapeutic direction for relapsed/refractory AML, it is associated with on-target toxicity. During the process of targeting and eliminating tumour cells, this therapy induces cytotoxicity against neutrophils, leading to severe granulocytopenia and significantly increasing the risk of infectious complications during CAR-T treatment, which has become a key bottleneck limiting its clinical application [152].

To address the severe granulocytopenia toxicity caused by CLL1 CAR-T therapy in AML, resulting from targeting CLL1 antigen on neutrophils, the research team designed novel modified CD15/CD16-CLL1 inhibitory CAR-T cells. Through structural optimization, CD16-CLL1 inhibitory CAR-T cells were screened, integrating PD1 and 2B4 blocking domains and replacing the CD16 scFv recognition domain with a single VHH fragment. These cells retain tumour-killing capacity while reducing neutrophil depletion. *In vivo* experiments confirmed their ability to effectively decrease the incidence of granulocytopenia. Follow-up plans include further clinical trial validation of this therapy's safety and efficacy [153], along with promoting its translation into universal CAR-T cells to meet clinical needs.

Aiming at the issues of autologous CAR-T cells in AML patients, including dominant effector phenotypes, high expression of exhaustion markers and impaired antitumour function, researchers constructed allogeneic CAR-T cells with knockout of HLA-I and TCR complex (HLA-IKO/TCRKO CAR-T) by combining CRISPR genome editing technology with the Sleeping Beauty transposon virus-free transfer system [154]. This protocol enables one-step generation of edited cells, with edited cells showing phenotypes similar to unedited CAR-T cells and consistent *in vivo* antitumour efficacy. Meanwhile, the technology features

safe vector integration, high editing specificity, and no obvious off-target effects. Large-scale production validation results indicate that these universal CAR-T cells meet clinical application requirements, providing a new technical pathway to address functional defects of autologous T cells in AML patients and overcome the bottleneck of allogeneic transplantation immune rejection.

5.2. CAR-NK cell therapy

CAR-NK cell therapy, as an emerging cellular immunotherapy, exhibits distinct advantages in the treatment of haematological malignancies including AML, such as milder GVHD, lower incidence of cytokine release syndrome, and broader antitumour effects compared with conventional CAR-T cell therapy. To address the inhibition of CAR-NK cells by HLA-E-NKG2A interaction in AML therapy, researchers used CRISPR/Cas9 technology to disrupt the *KLRC1* gene encoding NKG2A, constructing CD33-specific CAR-NK cells [155]. Single-cell multi-omics analysis showed favourable activation and maturation characteristics, and these cells exhibited enhanced cytotoxic activity against AML cells *in vivo*, confirming that this technology helps bypass AML immune suppression and provides a new direction for CAR-NK cell therapy in AML. Additionally, another study used baboon envelope pseudotyped lentiviral vectors to transduce primary haematopoietic NK cells, generating CD33-targeted CAR-modified natural killer cells (CD33-CAR-NK) [156]. *In vitro* experiments demonstrated stable CAR expression and unimpeded proliferation. In an OCI-AML2 xenograft mouse model, CD33-CAR-NK cells significantly reduced leukemic burden, prevented bone marrow infiltration by leukemic cells, and showed no obvious side effects, offering a novel therapeutic avenue for AML.

Aiming at the poor prognosis of relapsed/refractory AML and limited efficacy of CAR-T cells, studies explored the therapeutic effect of combining proteasome inhibitors with allogeneic CAR-expressing NK cells in AML [157]. Experiments confirmed that proteasome inhibitors sensitize AML cells to NK-mediated cytotoxicity by reducing class I HLA molecule expression, upregulating stress-related proteins, and enhancing NK cell killing activity – with synergistic effects notably improved when combined with CD33- and CD70-specific CARs. *In vivo* studies showed that this combination strategy inhibited AML growth and prolonged survival, providing a novel combinatorial approach for CAR-NK cell therapy in AML. However, head-to-head comparisons revealed that CAR-T cells outperform CAR-NK cells in CAR-mediated effector functions [158].

To address the challenges of relapse/refractoriness driven by LSCs in AML, as well as limitations of CAR-T cell therapy such as graft-versus-host disease risk and long autologous cell preparation cycles, researchers constructed third-generation anti-TIM3 CAR-modified CAR-TIM3 iPSCs using umbilical cord blood NK cell-derived iPSCs. These cells were differentiated into functional CAR-TIM3 NK cells under serum-free and feeder layer-free conditions [159]. The CAR-TIM3 NK cells exhibit significant cytotoxic activity against TIM3-positive AML cells with minimal impact on TIM3-negative cells. Their iPSC-based ‘off-the-shelf’ production system provides technical support for Good Manufacturing Practice-grade clinical translation.

Additionally, CAR-NK cell therapy has achieved critical breakthroughs in AML treatment: 1) TIM3-targeted CAR-NK cells: Leveraging expression differences of TIM3 between AML cells and HSPCs, these cells precisely recognize primary AML cells and reduce bone marrow and liver tumour burden *in vivo*, with controllable toxicity to HSPCs. CAR-TIM3 NK-92 cells downregulate surface TIM3 expression to minimize autologous killing, delay cell exhaustion, and enhance antitumour durability [160]. 2) CD33-targeted CAR-NK cells: Umbilical cord-derived NK cells modified by retroviral vectors have shown preclinical and phase I trial results indicating comparable killing efficacy against CD33⁺ AML cells to CAR-T cells, but with lower toxicity to haematopoietic stem cells. No severe immune effector cell-associated neurotoxicity syndrome or graft-versus-host disease occurred, with only low-grade cytokine release syndrome observed [161].

Both types of CAR-NK cells exhibit ‘off-the-shelf’ advantages (no need for individualized preparation, pre-storable for use), circumventing graft-versus-host disease risks of CAR-T cells and shortening preparation cycles, while demonstrating low off-target toxicity and high safety profiles. These features provide an innovative therapeutic direction for refractory AML. Future studies require expanding sample sizes to validate long-term efficacy and clinical translation feasibility.

5.3. Epigenetic regulation

Epigenetic regulatory drugs have demonstrated significant potential in CAR-T cell therapy for AML. Studies have shown that the histone deacetylase inhibitor valproic acid can synergistically enhance the cytotoxic activity of CAR-T cells (such as anti-CD123 CAR-T and CLL-1 CAR-T) against tumour cells by upregulating the expression of NKG2D ligands in AML cells, outperforming combination therapy with PD-1/PD-L1 antibodies [162]. *In vivo* experiments confirmed that combined treatment with valproic acid and CAR-T significantly prolonged mouse survival. This indicates that epigenetic modulators (such as valproic acid) as pretreatment strategies may break through AML resistance to CAR-T therapy by remodelling the tumour cell immunomicroenvironment, providing a new direction for epigenetic regulation-based combination therapies.

AML patients with relapse have poor prognosis, and CD70 has attracted significant attention as a potential therapeutic target. HMAs may upregulate CD70 through epigenetic regulation, providing preclinical evidence for the application of CD70-targeted CAR-T cell therapy in relapsed/refractory AML – particularly the therapeutic strategy of combining HMAs to enhance target exposure [163]. The self-renewal property of LSCs is a key factor in poor AML prognosis. Studies have shown that IL-21 secreted by CD4⁺ T cells acts as a negative regulator of LSCs self-renewal, inducing asymmetric division and differentiation of LSCs *via* activating the p38-MAPK signalling pathway, which significantly reduces LSCs numbers and prolongs survival in mouse models [164]. Clinical data indicate that serum IL-21 levels at diagnosis are an independent positive biomarker for AML patient prognosis, closely associated with improved survival rates and complete remission rates after high-dose chemotherapy. *In vivo* experiments confirm that IL-21 pretreatment not only inhibits LSCs function but also enhances the cytotoxic effects of cytarabine and CD70 CAR-T cells against LSCs.

In KMT2A-rearranged AML, activating mutations in FLT3 and RAS (FLT3-ITD, FLT3-N676K, NRAS-G12D) reshape the chromatin accessibility landscape, leading to epigenetic alterations that combine shared core features with unique characteristics. These changes are driven by distinct transcription factor networks including AP-1, Runx1 and Stat5a/Stat5b, which orchestrate divergent transcriptional programs [165]. Specifically, FLT3-N676K and NRAS-G12D activate immune-related transcriptional programs, promoting expression of NKG2D ligands to enhance sensitivity to CAR-T cell-mediated killing. The histone deacetylase inhibitor LBH589 further upregulates NKG2D ligand levels, potentiating combinatorial cytotoxic effects. This provides a novel direction for integrating CAR-T cell therapy with epigenetic regulation to tackle KMT2A-rearranged AML.

As a potential target for AML, CD70-targeted traditional CAR-T cells (using scFv or truncated CD27) are limited by issues such as cell exhaustion and immunogenicity. Research has developed nanobody-targeted nb70CAR-T cells, which exhibit cytotoxicity against CD70-positive AML cells. However, CD70 expression is insufficient in primary AML blasts. Decitabine and chidamide can upregulate CD70 expression, enhancing the efficacy of nb70CAR-T cells. These findings suggest that combining nanobody-based CAR-T therapy with epigenetic modulators may represent a novel therapeutic strategy for AML [166].

5.4. $\gamma\delta$ T cell therapy for AML

To address the challenge of targeted therapy in myeloid malignancies (such as AML) due to the lack of specific surface antigens for traditional CAR-T cells, Branella et al. developed an engineered $\gamma\delta$ T cell therapy targeting c-kit *via* ligand [167]. c-kit is highly expressed in AML and associated with chemoresistance, while $\gamma\delta$ T cells can be safely regulated through transient transfection systems, enabling allogeneic 'off-the-shelf' therapy. *In vivo* studies showed prolonged survival in AML-transplanted mice, though limited by bone marrow homing ability. This research provides preclinical evidence for ligand-targeted immunotherapy in AML, supporting further development of SCF-based $\gamma\delta$ T cell therapy.

6. Conclusion

CAR-T cell therapy has brought revolutionary breakthroughs to AML treatment. By genetically engineering T cells to achieve precise tumour targeting, it demonstrates significant potential in overcoming the

Table 2. List of abbreviations (according to alphabetical order).

Abbreviations	Full name
4-1BB	Tumour Necrosis Factor Receptor Superfamily Member 9
AbTCR	antibody-T cell receptor
AdCAR	adapter CAR
ADCC	antibody-dependent cell-mediated cytotoxicity
ADGRE2	adhesion G protein-coupled receptor E2
ADGRG1	adhesion G protein-coupled receptor G1
AKT	Protein Kinase B
ALC	absolute lymphocyte count
ALL	acute lymphoblastic leukaemia
allo-HSCT	allogeneic haematopoietic stem cell transplantation
AMkL	Acute megakaryoblastic leukaemia
AML	Acute Myeloid Leukaemia
AP-1	activator protein 1
ATP	adenosine triphosphate
BATF	Basic Leucine Zipper Transcription Factor, ATF-Like
BCL-2	B-cell lymphoma/leukaemia-2 gene
BEs	base editors
BRD4	Bromodomain-containing protein 4
C/EBP	CCAAT/enhancer-binding protein
C-JUN	cellular proto-oncogene c-jun
CAR	Chimeric antigen receptor
CARKR	CAR with lysine-to-arginine mutation
CAR-T	Chimeric antigen receptor T cell
CD86	cluster of differentiation 86
CIK	cytokine-induced killer cells
CLEC9A	C - type Lectin Domain Family 9 Member A
CLEC12A	C-type Lectin Domain Family 12 Member A
CLL1	C-type lectin-like molecule 1
CRP	C-Reactive Protein
CRS	Cytokine Release Syndrome
CSF1R	colony-stimulating factor 1 receptor
CSR	costimulatory signal receptor
CTLA-4	cytotoxic T-lymphocyte-associated protein 4
CXCR4	C-X-C chemokine receptor type 4
DNMT3A	DNA Methyltransferase 3A
dNPM1	mutated Nucleophosmin 1
dVHH	domain of variable heavy chain of heavy-chain antibody
eGFP	Enhanced Green Fluorescent Protein
EGR1	Early Growth Response 1
EMDs	extramedullary diseases
ERK	extracellular signal-regulated kinase
FAS	Fas cell surface death receptor
FLT3	Fms-like Tyrosine Kinase 3
FLT3-CART	FLT3-targeted CAR-T cells
FRET	fluorescence resonance energy transfer
G-CSF	Granulocyte Colony-Stimulating Factor
GLUT	glucose transporter
GM-CSF	Granulocyte-Macrophage Colony-Stimulating Factor
GRP78	glucose-regulated protein 78
GVHD	graft-versus-host disease
HER2	Human Epidermal Growth Factor Receptor 2
HIF-1 α	hypoxia-inducible factor-1 α
HLA-I	Human Leukocyte Antigen Class I
HMAAs	Hypomethylating agents
HSCs	haematopoietic stem cells
HSPCs	haematopoietic stem progenitor cells
ICAM-1	Intercellular Adhesion Molecule 1
ICANS	Immune Effector Cell-Associated Neurotoxicity Syndrome
IFN- γ	γ -interferon
IL	Interleukin
IL13R α 2	Interleukin-13 Receptor Alpha 2
IL1RAP	Interleukin 1 Receptor Accessory Protein
iPSCs	induced pluripotent stem cells
KMT2A	lysine methyltransferase 2A
IRF	Interferon Regulatory Factor
LAG3	Lymphocyte Activation Gene 3
LILRB3	leukocyte immunoglobulin-like receptor B3
LSCs	leukaemia stem cells
M-MDSCs	Myeloid-derived suppressor cells
MAPK	mitogen-activated protein kinase
MIP-1 β	Macrophage Inflammatory Protein-1 β
MM	Multiple myeloma
MRD	minimal residual disease

(Continued)

Table 2. Continued.

Abbreviations	Full name
NFAT	Nuclear Factor of Activated T cells
NF- κ B	Nuclear Factor kappa-light-chain-enhancer of activated B cells
NK	natural killer
NKG2D	Natural Killer Group 2, Member D
NRAS	neuroblastoma RAS viral oncogene homologue
NSCLC	non-small cell lung cancer
PD-1	Programmed Cell Death Protein 1
PD-L1	programmed death ligand 1
PEs	prime editors
PET	positron emission tomography
PRAME	Preferentially Expressed Antigen in Melanoma
PU.1	PUR transcription factor
RAS	rat sarcoma virus oncogene homologue
Rb	Retinoblastoma protein
ROR1	receptor tyrosine kinase-like orphan receptor 1
Runx1	runt-related transcription factor 1
SCF	stem cell factor
scFv	single-chain variable fragment
SdAb	single-domain antibody
SHP2	Src homology 2 domain-containing protein tyrosine phosphatase 2
SLC2A1	solute carrier family 2 member 1
Src	Src family tyrosine kinase
Stat	Signal transducer and activator of transcription
T-ALL	T-cell acute lymphoblastic leukaemia
TCF-1	T Cell Factor 1
TCR	T Cell Receptor
TFAM	transcription factor A
TGF- β	Transforming Growth Factor- β
TIM3	T cell immunoglobulin and mucin domain 3
TNF- α	Tumour Necrosis Factor- α
Tregs	regulatory T cells
UCAR-T	universal CAR-T

limitations of traditional chemotherapy and haematopoietic stem cell transplantation. Current clinical studies targeting CD33 [75], CD123 [77–79], NKG2D [76], CLL-1 [80, 82, 83, 84] and LILRB3 [74] have preliminarily validated the safety and antitumour activity of CAR-T therapy. These 11 representative studies, covering preclinical mechanistic research and early clinical trials, fully demonstrate the translational potential of diverse CAR-T targets for AML, despite the gap between preclinical efficacy and clinical success. Innovations in dual-target design, gene editing technologies and immunomicroenvironment regulation also provide new strategies to address core challenges such as on-target toxicity, antigen escape and insufficient cell persistence. However, a major limitation of this review is that it extensively summarizes preclinical studies and mechanistic insights, while clinical aspects, including large-scale clinical trials, long-term clinical efficacy, clinical safety monitoring and real-world clinical practice, are insufficiently addressed. It is critical to recognize that impressive preclinical results do not always translate into clinical success, which restricts the direct guidance of conclusions to clinical practice.

Beyond these limitations, clinical translation of CAR-T therapy for AML still faces multiple challenges: ‘off-target effects’ due to insufficient tumour antigen specificity, suppression of CAR-T cell function by the immunosuppressive microenvironment, and limited cell expansion and persistence – all of which require further technological iterations to overcome. In addition, optimization of T-cell collection timing, reduction of chemotherapy-induced T-cell damage, and rational formulation of CAR-T cell allocation and reinfusion protocols are equally critical for improving therapeutic outcomes. Despite encouraging preclinical and early clinical results, many current strategies remain at the experimental stage and long-term safety, clonal immune escape, extramedullary recurrence and cost-effectiveness still lack sufficient clinical validation. The dynamic and spatial heterogeneity of the AML microenvironment, as well as metabolic immunosuppression, further increase the uncertainty of therapeutic efficacy. Future research should strengthen multi-centre clinical trials, long-term follow-up and mechanistic validation, focusing on the balance between efficacy and safety. In the future, integration of emerging technologies such as UCAR-T, CAR-NK cell therapy and epigenetic regulation is expected to drive AML treatment towards more precise and safer directions. Meanwhile, multidimensional strategies combining immune checkpoint inhibitors,

optimizing target combinations and remodelling the tumour microenvironment may become key pathways to breakthrough existing bottlenecks.

Overall, the preclinical application of CAR-T cell therapy in AML has demonstrated broad prospects, but its standardized clinical implementation and technological innovation still require deep integration of basic research and clinical trials to ultimately achieve the goal of precision immunotherapy and bring more survival hope to patients with relapsed/refractory AML. (The complete abbreviations are shown in Table 2.)

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Data Availability statement

The data that support the findings of this review are available from the corresponding author upon reasonable request.

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